Chapter 8. Cluster Sampling

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1 Introduction

Suppose we want the weekly food expenditure of families in Seoul. SRS, stratified or systematic sampling of families requires a list of all families (rarely available), and the sampled families would be scattered all over the city, which makes interviewing expensive. A list of city blocks, however, is easy to obtain, and interviewing every family in a few blocks is cheap.

Definition 8.1

A **cluster sample** is a simple random sample in which each sampling unit is a collection (cluster) of elements; every element of a selected cluster is observed.

Cluster sampling is effective when

- a good frame of elements is unavailable or costly, while a frame of clusters is easy to obtain;
- the cost of observation increases with the distance between elements.

Since it is SRS of clusters, the estimators parallel those of Ch. 2, with cluster totals playing the role of observations.

Notation.

N	number of clusters in the population
n	number of clusters selected by SRS
m_i	number of elements in cluster i, $i = 1, \ldots, N$
$\bar{m} = \frac{1}{n}\sum_{i=1}^{n} m_i$	average cluster size in the sample
$M = \sum_{i=1}^{N} m_i$	number of elements in the population
$\bar{M} = M/N$	average cluster size in the population
y_{ij}	value of the jth element in cluster i
$y_i = \sum_{j=1}^{m_i} y_{ij}$	total of cluster i

2 Estimation of the Population Mean

The population mean per element is

$$\mu = \frac{\sum_{i=1}^{N}\sum_{j=1}^{m_i} y_{ij}}{\sum_{i=1}^{N} m_i} = \frac{\sum_{i=1}^{N} y_i}{M},$$

a ratio of the cluster totals y_i to the cluster sizes m_i. Its natural estimator is therefore a ratio estimator (Ch. 5) with $x_i = m_i$.

Theorem 8.2: Cluster estimator of μ

$$\bar{y}_{\text{cl}} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}\, s_c^2, \qquad s_c^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_{\text{cl}} m_i)^2,$$

where $\bar{M}$ is replaced by $\bar{m}$ if M is unknown. An approximate 95% interval is $\bar{y}_{\text{cl}} \pm 2\sqrt{\widehat{\text{Var}}(\bar{y}_{\text{cl}})}$.

Derivation: from the ratio estimator

With $x_i = m_i$, $R = \mu$ and $\mu_x = \bar{M}$, Ch. 5 gives

$$\text{Var}(\bar{y}_{\text{cl}}) \approx \frac{1}{\bar{M}^2}\cdot\frac{N-n}{N}\cdot\frac{1}{n}\cdot\frac{1}{N-1}\sum_{i=1}^{N}(y_i - \mu m_i)^2 = \frac{1}{\bar{M}^2}\frac{N-n}{Nn}(S_y^2 + \mu^2 S_m^2 - 2\mu\rho S_m S_y),$$

and replacing population quantities by sample ones yields $\widehat{\text{Var}}(\bar{y}_{\text{cl}})$ and the alternative form

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}(s_y^2 + \bar{y}_{\text{cl}}^2 s_m^2 - 2\bar{y}_{\text{cl}}\hat{\rho} s_m s_y), \qquad s_m^2 = \frac{1}{n-1}\sum_{i=1}^{n}(m_i - \bar{m})^2,$$

where s_y^2 is the sample variance of the cluster totals. Cluster totals nearly proportional to cluster sizes ($y_i \approx \mu m_i$) give a small variance.

2.1 A regression-type estimator

Treating the cluster size as the covariate, $y_i = \beta_0 + \beta_1 m_i + \varepsilon_i = \beta_0' + \beta_1(m_i - \bar{m}) + \varepsilon_i$, the regression estimator of the mean cluster total is

$$\hat{\mu}_t = \bar{y}_t + \hat{\beta}_1(\bar{M} - \bar{m}), \qquad \hat{\beta}_1 = \frac{\sum(m_i - \bar{m})(y_i - \bar{y}_t)}{\sum(m_i - \bar{m})^2}, \quad \bar{y}_t = \frac{1}{n}\sum y_i,$$

with $\widehat{\text{Var}}(\hat{\mu}_t) = \frac{N-n}{Nn} M S_E$ as in Ch. 7. Dividing by $\bar{M}$ gives an estimator of μ. It beats the ratio form $\bar{y}_{\text{cl}}$ unless the LS line of y_i on m_i passes through the origin.

3 Estimation of the Population Total

3.1 M known

Since $\tau = M\mu$,

$$\hat{\tau} = M\bar{y}_{\text{cl}} = M\frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\text{Var}}(\hat{\tau}) = M^2\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = N^2\frac{N-n}{Nn}s_c^2.$$

3.2 M unknown

Often the number of elements is unknown. Then use the mean of the cluster totals,

$$\bar{y}_t = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \text{E}(\bar{y}_t) = \frac{1}{N}\sum_{i=1}^{N} y_i = \frac{\tau}{N},$$

which is unbiased for the average cluster total, so $N\bar{y}_t$ is unbiased for τ:

$$\hat{\tau} = N\bar{y}_t = \frac{N}{n}\sum_{i=1}^{n} y_i, \qquad \widehat{\text{Var}}(\hat{\tau}) = N^2\frac{N-n}{Nn}s_t^2, \qquad s_t^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_t)^2.$$

This is SRS of the cluster totals and ignores the sizes m_i; it is less precise than $M\bar{y}_{\text{cl}}$ when the totals vary mainly because the sizes vary.

If all $m_i = m$, then $M = Nm$, $\bar{y}_{\text{cl}} = \bar{y}_t/m$, and the two estimators of τ coincide.

4 Comparison with Simple Random Sampling

Assume equal cluster sizes $m_i = m$, so $M = Nm$ and

$$\bar{y}_{\text{cl}} = \frac{\sum_{i=1}^{n} y_i}{nm} = \frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} y_{ij} = \frac{\bar{y}_t}{m}.$$

Let $\bar{y}_i = y_i/m$ be the mean of cluster i. Then

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{1}{m^2}\widehat{\text{Var}}(\bar{y}_t) = \frac{1}{m^2}\frac{N-n}{Nn}\frac{1}{n-1}\sum_i (m\bar{y}_i - m\bar{y}_{\text{cl}})^2 = \frac{N-n}{Nn}\frac{1}{n-1}\sum_{i=1}^{n}(\bar{y}_i - \bar{y}_{\text{cl}})^2.$$

Treat the nm observed elements as an $m \times n$ table with column means $\bar{y}_i$ and grand mean $\bar{y}_{\text{cl}}$:

$$\underbrace{\sum_{i=1}^{n}\sum_{j=1}^{m}(y_{ij} - \bar{y}_{\text{cl}})^2}_{SST} = \underbrace{\sum_i\sum_j (y_{ij} - \bar{y}_i)^2}_{SSW} + \underbrace{m\sum_i (\bar{y}_i - \bar{y}_{\text{cl}})^2}_{SSB},$$

with $MSB = SSB/(n-1)$ and $MSW = SSW/[n(m-1)]$. Then

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nnm}\,MSB.$$

If the same nm elements had been drawn by SRS from the Nm elements, with sample variance s^2,

$$\widehat{\text{Var}}(\bar{y}) = \frac{Nm - nm}{Nm}\cdot\frac{s^2}{nm} = \frac{N-n}{Nnm}\,s^2.$$

Remark 8.3: Efficiency of cluster sampling

The relative efficiency of cluster sampling versus SRS is measured by comparing MSB with s^2. Cluster sampling is more efficient when there is little variation between clusters, i.e. when each cluster is a miniature of the whole population (heterogeneous within). In most real populations, neighboring elements are alike, $MSB > s^2$, and cluster sampling is less efficient per element than SRS; it is chosen for its lower cost per element.

5 Selecting the Sample Size

Setting $B = 2\sqrt{\frac{N-n}{nN\bar{M}^2}s_c^2}$ and solving as in Ch. 2:

- for μ with $\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND + s_c^2}$, $D = \dfrac{B^2\bar{M}^2}{4}$;
- for τ with $M\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND + s_c^2}$, $D = \dfrac{B^2}{4N^2}$;
- for τ with $N\bar{y}_t$: $n = \dfrac{Ns_t^2}{ND + s_t^2}$, $D = \dfrac{B^2}{4N^2}$,

with s_c^2, s_t^2 taken from a previous survey or a pilot sample.

6 Estimation of a Population Proportion

Let a_i be the number of elements in cluster i having the characteristic. Then p is the ratio $\sum a_i / \sum m_i$ over the population, and

$$\hat{p}_{\text{cl}} = \frac{\sum_{i=1}^n a_i}{\sum_{i=1}^n m_i}, \qquad \widehat{\text{Var}}(\hat{p}_{\text{cl}}) = \frac{N-n}{nN\bar{M}^2}s_p^2, \qquad s_p^2 = \frac{1}{n-1}\sum_{i=1}^n (a_i - \hat{p}_{\text{cl}} m_i)^2,$$

with $\bar{M} \approx \bar{m}$ if M is unknown. The sample size for a bound B is $n = Ns_p^2/(ND + s_p^2)$ with $D = B^2\bar{M}^2/4$.

7 Cluster Sampling Combined with Stratification

To survey the number of books read by 4th and 5th graders, divide the population into two strata (grades) and take a cluster sample (of classes) in each. In stratum ℓ let N_ℓ be the number of clusters, $\bar{y}_{t\ell}$ the mean cluster total and $\bar{m}_\ell$ the mean cluster size in the sample. Then

$$\text{average cluster total: } \frac{1}{N}(N_1\bar{y}_{t1} + N_2\bar{y}_{t2}), \qquad \text{average cluster size: } \frac{1}{N}(N_1\bar{m}_1 + N_2\bar{m}_2),$$

$$\bar{y}_c = \frac{N_1\bar{y}_{t1} + N_2\bar{y}_{t2}}{N_1\bar{m}_1 + N_2\bar{m}_2} = \frac{\sum_\ell N_\ell \bar{y}_{t\ell}}{\sum_\ell N_\ell \bar{m}_\ell},$$

a combined ratio estimator. Its estimated variance is

$$\widehat{\text{Var}}(\bar{y}_c) = \frac{1}{\left(\sum_\ell N_\ell \bar{m}_\ell\right)^2}\sum_{\ell=1}^L N_\ell(N_\ell - n_\ell)\frac{s_{t\ell}^2}{n_\ell},$$

where $s_{t\ell}^2 = \frac{1}{n_\ell - 1}\sum_{i=1}^{n_\ell}(y_i - \bar{y}_{t\ell})^2$ is the sample variance of the cluster totals in stratum ℓ. The numerator is $\sum_\ell N_\ell^2 \widehat{\text{Var}}(\bar{y}_{t\ell})$, the estimated variance of the stratified estimate of the population total, and the denominator is the squared estimated number of elements.

8 Cluster Sampling with Probabilities Proportional to Size

8.1 The PPS estimator

Let the population consist of units with values $y_1, \dots, y_N$ and suppose sampling is with replacement, with z_i the probability that unit i is selected on any one draw ($\sum z_i = 1$).

Theorem 8.4: Hansen–Hurwitz estimator

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{z_i}$$

is unbiased for τ for any choice of $z_i > 0$, with $\widehat{\text{Var}}(\hat{\tau}_{\text{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}\left(\frac{y_i}{z_i} - \hat{\tau}_{\text{pps}}\right)^2$.

Proof. Each draw gives an iid value $t = y_I/z_I$ with $P(I = k) = z_k$, so $\text{E}(t) = \sum_k z_k \frac{y_k}{z_k} = \tau$. The average of n iid unbiased values is unbiased, and its variance is estimated by s_t^2/n. ■

With $z_i = 1/N$ (equal probability), $\hat{\tau}_{\text{pps}} = \frac{N}{n}\sum y_i = N\bar{y}$, the usual estimator. The variance is small when $z_i \propto y_i$, so we want z_i roughly proportional to y_i.

8.2 Application to clusters

Cluster totals tend to be proportional to cluster sizes, so choose $z_i = m_i/M$. With y_i the cluster total and $\bar{y}_i = y_i/m_i$ the cluster mean,

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{m_i/M} = \frac{M}{n}\sum_{i=1}^{n}\bar{y}_i, \qquad \hat{\mu}_{\text{pps}} = \frac{\hat{\tau}_{\text{pps}}}{M} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i :$$

the estimator of μ is the plain average of the sampled cluster means.

Theorem 8.5

$$\widehat{\text{Var}}(\hat{\tau}_{\text{pps}}) = \frac{M^2}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\text{pps}})^2, \qquad \widehat{\text{Var}}(\hat{\mu}_{\text{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\text{pps}})^2.$$

Proof. The $\bar{y}_i$ are iid draws (with replacement), so $\widehat{\text{Var}}(\frac{1}{n}\sum\bar{y}_i) = \frac{1}{n}\cdot\frac{1}{n-1}\sum(\bar{y}_i - \hat{\mu}_{\text{pps}})^2$; multiply by M^2 for the total. ■

Remark 8.6: Equal sizes

If $m_i = m$, then $\hat{\mu}_{\text{pps}} = \frac{1}{n}\sum_i\frac{1}{m}\sum_j y_{ij} = \bar{y}_{\text{cl}} = \bar{y}_t/m$, and since $s_t^2 = \frac{m^2}{n-1}\sum(\bar{y}_i - \bar{y}_{\text{cl}})^2$,

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{1}{m^2}\frac{N-n}{Nn}s_t^2 = \frac{N-n}{N}\cdot\frac{1}{n(n-1)}\sum(\bar{y}_i - \bar{y}_{\text{cl}})^2,$$

which is $\widehat{\text{Var}}(\hat{\mu}_{\text{pps}})$ up to the fpc (PPS draws are with replacement).

9 Selected exercises

Motivation. The comparison with SRS above was made through estimated variances. The first item gives the exact population version, the design effect $1 + (m-1)\rho$; the second derives the exact

variance of the PPS estimator and shows why selecting clusters proportional to size removes the variation of cluster sizes from the error.

Exercise 8.1 (Design effect of cluster sampling)

Let all N clusters have m elements ($M = Nm$), let μ and $S^2 = \frac{1}{Nm-1}\sum_{i=1}^{N}\sum_{j=1}^{m}(y_{ij}-\mu)^2$ be the element mean and variance, and define the intracluster correlation

$$\rho = \frac{\sum_{i=1}^{N}\sum_{j\neq j'}(y_{ij}-\mu)(y_{ij'}-\mu)}{(m-1)(Nm-1)S^2}.$$

For an SRS of n clusters, $f = n/N$, show that

$$\operatorname{Var}(\bar{y}_{\mathrm{cl}}) = \frac{1-f}{nm}\,S^2 \cdot \frac{Nm-1}{m(N-1)}\,[1+(m-1)\rho],$$

and that the ratio of $\operatorname{Var}(\bar{y}_{\mathrm{cl}})$ to the variance of the mean of an SRS of nm elements is approximately $1+(m-1)\rho$ when N is large.

Solution. [Supplement]

Step 1. With equal sizes, $\bar{y}_{\mathrm{cl}} = \frac{1}{n}\sum_{i\le n}\bar{y}_i$ is the SRS mean of the N cluster means $\bar{y}_i = \frac{1}{m}\sum_j y_{ij}$, whose population mean is μ. By Ch. 2,

$$\operatorname{Var}(\bar{y}_{\mathrm{cl}}) = \frac{1-f}{n}S_b^2, \qquad S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\bar{y}_i-\mu)^2.$$

Step 2. Expanding a square of a sum,

$$\begin{aligned}\sum_{i=1}^{N}(\bar{y}_i-\mu)^2 &= \frac{1}{m^2}\sum_i\Big(\sum_j(y_{ij}-\mu)\Big)^2 = \frac{1}{m^2}\Big[\sum_i\sum_j(y_{ij}-\mu)^2 + \sum_i\sum_{j\neq j'}(y_{ij}-\mu)(y_{ij'}-\mu)\Big]\\ &= \frac{1}{m^2}\Big[(Nm-1)S^2 + (m-1)(Nm-1)S^2\rho\Big] = \frac{(Nm-1)S^2}{m^2}[1+(m-1)\rho].\end{aligned}$$

Step 3. Substituting into Step 1 gives the stated formula.

Step 4. (Design effect.) An SRS of nm of the Nm elements has sampling fraction $nm/(Nm) = f$ and variance $\frac{1-f}{nm}S^2$. The ratio is $\frac{Nm-1}{m(N-1)}[1+(m-1)\rho]$, and $\frac{Nm-1}{m(N-1)} \to 1$ as $N\to\infty$.

When elements of a cluster are alike ($\rho > 0$), a cluster sample of nm elements is worth only about $nm/[1+(m-1)\rho]$ independently drawn elements; even $\rho = 0.05$ doubles the variance for clusters of size $m = 21$. Cluster sampling is efficient per element only if $\rho \le 0$, matching the remark on efficiency above. ∎

Exercise 8.2 (Exact variance of the PPS estimator)

In the setting of the Hansen–Hurwitz theorem (draws with replacement, selection probabilities $z_k > 0$, $\sum z_k = 1$), show that

(a) $\operatorname{Var}(\hat{\tau}_{\mathrm{pps}}) = \dfrac{1}{n}\sum_{k=1}^{N} z_k\Big(\dfrac{y_k}{z_k}-\tau\Big)^2$;

(b) if all $y_k > 0$, the variance is zero iff $z_k = y_k/\tau$ for all k;

(c) $\widehat{\operatorname{Var}}(\hat{\tau}_{\mathrm{pps}})$ of the theorem is unbiased for $\operatorname{Var}(\hat{\tau}_{\mathrm{pps}})$;

(d) for clusters with $z_i = m_i/M$, $\operatorname{Var}(\hat{\tau}_{\mathrm{pps}}) = \dfrac{M}{n}\sum_{i=1}^{N} m_i(\bar{y}_i-\mu)^2$, where $\bar{y}_i$ is the mean of cluster i.

Solution. [Supplement] Let $t_r = y_{I_r}/z_{I_r}$, $r = 1, \ldots, n$, where $I_1, \ldots, I_n$ are the independent draws, $P(I_r = k) = z_k$. Then $\hat{\tau}_{\text{pps}} = \bar{t}$.

(a) The t_r are iid with mean τ (proof of the theorem) and variance $V_1 = \text{E}(t_1 - \tau)^2 = \sum_k z_k(y_k/z_k - \tau)^2$. Hence $\text{Var}(\bar{t}) = V_1/n$.

(b) V_1 is a sum of nonnegative terms with positive weights z_k, so $V_1 = 0$ iff $y_k/z_k = \tau$ for every k, i.e. $z_k = y_k/\tau$; these z_k are positive and sum to 1. Such a design is impossible in practice (it needs the y_k), but it shows that the variance is small when z_k is nearly proportional to y_k.

(c) For iid t_r the sample variance $s_t^2 = \frac{1}{n-1}\sum_r (t_r - \bar{t})^2$ is unbiased for V_1 (the with-replacement case of Ch. 2). Hence $\text{E}[s_t^2/n] = V_1/n = \text{Var}(\hat{\tau}_{\text{pps}})$, and s_t^2/n is the $\widehat{\text{Var}}(\hat{\tau}_{\text{pps}})$ of the theorem.

(d) With y_i the cluster total, $y_i/z_i = My_i/m_i = M\bar{y}_i$ and $\tau = M\mu$. By (a),

$$\text{Var}(\hat{\tau}_{\text{pps}}) = \frac{1}{n}\sum_i \frac{m_i}{M} M^2(\bar{y}_i - \mu)^2 = \frac{M}{n}\sum_i m_i(\bar{y}_i - \mu)^2.$$

Only the variation of the cluster *means* enters. By contrast, $N\bar{y}_t$ (equal probabilities) has variance driven by the cluster *totals* $y_i = m_i\bar{y}_i$, which vary also because the sizes m_i vary. ■